

MDHGNN Project Description

Weekly Frequency Prediction for Copper Futures Based on Yield Modeling

I. Project Overview

This project implements the multi-scale dynamic hypergraph neural network MDHGNN for weekly price forecasting of copper futures. The model's primary prediction objective is future returns, with price forecasts derived from previous prices and predicted returns; directional accuracy is determined by analyzing return signs.

bore	The Meaning of the Project
Directly predict the target	Future h-week return vector.
price expectation	Use the previous price P_{t-1} and the predicted return to reconstruct future prices, then calculate the price error.
Direction Prediction	Do not train the classifier separately; the direction is determined by whether the predicted return is greater than 0.
Main Frequency and Step Size	The frequency experiment primarily reports results for $h=1$ and $h=2$.

II. Data and Sample Period

To ensure all variables represent genuine historical observations, the experiment exclusively utilized the price range covered by copper futures, WTI crude oil, gold, silver, and the US Dollar Index. Since the earliest available silver futures data date back to July 22,2011, the model training dataset excluded data prior to that year.

project	explain
Common True Daily Frequency Range	From July 22,2011 to January 29,2016, there are a total of 1,146 rows. The starting point is determined by the earliest available date for Baiyin Futures.
After frequency-based resampling	From July 22,2011 to January 29,2016, there were a total of 237 rows.
After adding technical indicators	From January 13,2012 to January 29,2016, there were a total of 212 rows.
Feature Count	Thirty indicators, including copper futures OHLCV, technical indicators, and external market variables.
Window Length	Frequency experiment: $W = 12$.

data partitioning

gather	Search Index	Date Range	line number	Actual Target Sample
training set	0-147	January 13,2012 to November 7, 2014	148	136, h=1
validation set	148-168	November 14,2014 to April 3, 2015	21	21, h=1
test set	169-211	April 10,2015 to January 29,2016	43	42, h=1

When $h=2$, due to the increased prediction step size, the actual number of target samples in the training set and test set is 135 and 41, respectively.

Prevent data leakage

1. Training, validation, and testing datasets should be divided in chronological order; avoid random partitioning across time periods.
2. Z-score normalization uses only the mean and standard deviation of the training set, then applies them to the validation and test sets.
3. The input window for the t -th sample contains only $x_{\{t-W\}}, \dots, x_{\{t-1\}}$ and does not include the target-period price.
4. External variables are aligned by date and resampled to weekly frequency; data prior to July 22,2011 are excluded from the modeling sample.

III. Prediction Objectives

$$X_t = [x_{\{t-W\}}, \dots, x_{\{t-1\}}]$$

$$r_{\{t,j\}} = (P_{\{t+j-1\}} - P_{\{t-1\}}) / P_{\{t-1\}}, j = 1, \dots, h$$

$$\hat{P}_{\{t+j-1\}} = P_{\{t-1\}} * (1 + \hat{r}_{\{t,j\}})$$

Therefore, when $h=1$, the target is $(P_t - P_{\{t-1\}}) / P_{\{t-1\}}$; for $h=2$, the second price adjustment step uses $P_{\{t+1\}}$.

IV. Model Structure

MDHGNN comprises two pathways: the temporal pathway employs linear projection, position encoding, GRU, and attention pooling; the structural pathway treats input features as hypergraph nodes and learns higher-order relationships between variables through feature hypergraph convolution. The two pathways are fused via learnable weights, and the final output is the return rate generated by the prediction head.

module	explain
Time Path	Learn temporal dependencies from the input window, with the core components being GRUs and attention pooling.
Structure Path	Use 30 input features as nodes and employ hyperedges to represent higher-order relationships between features.

Integration Layer	Use learnable weights to combine temporal representations and structural representations.
Prediction Head	Output the future yield at step h.

Hyperedge Type

type	Construction Method	quantity
Correlation Soft Super Edge	Generate soft associations based on feature similarity in the input window.	90
Domain Prior Hyper-edge	It is established based on economic indicators such as price, trend, momentum, volatility, trading volume, external markets, and overall market conditions.	9
Adaptive Hyperedge	It is generated through learnable projections based on feature node embeddings.	8
amount to	The node has 30 input features.	107

V. Baseline Methods and Metrics

method	explain
Naive persistence	With a parameter-free persistent baseline, the predicted return is always 0, indicating that the price in the next period equals that of the previous period.
GRU	Two-layer GRU regression model.
MLP	After expanding the input window, use a multi-layer fully connected network for regression.
Transformer	Transformer Encoder Regression Baseline.

The evaluation metrics include price-related indicators such as RMSE, MAE, MAPE, and R^2 ; return-related metrics like Ret RMSE and Ret R^2 ; directional indicators including DA, DA_{up}, and DA_{down}; and strategy-based metrics such as Sharpe ratio, Annual Return, and Maximum Drawdown.

Ret RMSE Formula

The model predicts returns in decimal form, not percentage returns. The single-step return is defined as:

$$r_t = (P_t - P_{t-1}) / P_{t-1}$$

Therefore, the RMSE of the yield prediction error is first calculated in decimal yield units and then multiplied by 10^4 to convert it to basis points.

$$\text{Ret_RMSE(bp)} = 10^4 * \sqrt{(1/N) * \sum_{i=1}^N (r_i - \hat{r}_i)^2}$$

For multi-step forecasting, Ret_RMSE is expressed as:

$$\text{Ret_RMSE}(\text{bp}) = 10^4 * \sqrt{\frac{1}{N \cdot h}} * \sum_{i=1}^N \sum_{j=1}^h (r_{i,j} - \hat{r}_{i,j})^2$$

Here, r represents the true decimal return rate, $\hat{r}$ denotes the predicted decimal return rate, N indicates the number of test samples, and h stands for the prediction step size. Multiplying by 10^4 is done because $1 \text{ bp} = 0.01\% = 0.0001$.

Formula References

use	References
Yield Definition	Campbell, J. Y., Lo, A. W., and MacKinlay, A. C. (1997). The Econometrics of Financial Markets. Princeton University Press.
RMSE Prediction Error Indicator	Hyndman, R. J., and Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679-688. DOI: 10.1016/j.ijforecast.2006.03.001.
Base Point Conversion	Hull, J. C. Hull, J. C. (2018). Options, Futures, and Other Derivatives. 10th edition. Pearson. Basis points are commonly used to express interest-rate and return differences, where $1 \text{ bp} = 0.01\% = 0.0001$.

VI. Main Experimental Results

The following results were regenerated using the current code on a shared real sample set with frequency parameters and random seeds of 42,123, and 456.

h=1 Results

method	RMSE		MAE		MAPE	R2		DA
MDHGNN	0.0685 0.0018	±	0.0537 0.0017	±	2.25 ± 0.07	0.9366 0.0032	±	48.4 ± 11.4
GRU	0.0815 0.0109	±	0.0657 0.0106	±	2.75 ± 0.46	0.9087 0.0252	±	43.7 ± 11.2
MLP	0.0748 0.0049	±	0.0598 0.0055	±	2.50 ± 0.24	0.9242 0.0101	±	44.4 ± 14.8
Transformer	0.0869 0.0215	±	0.0698 0.0188	±	2.87 ± 0.71	0.8919 0.0538	±	56.3 ± 11.2
Naive persistence	0.0685 0.0000	±	0.0533 0.0000	±	2.24 ± 0.00	0.9365 0.0000	±	64.3 ± 0.0

h=2 Results

method	RMSE		MAE		MAPE	R2		DA
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MDHGNN	0.1034 ± 0.0097	0.0846 ± 0.0095	3.55 ± 0.44	0.8527 ± 0.0283	50.8 ± 2.1
GRU	0.1301 ± 0.0271	0.1084 ± 0.0279	4.62 ± 1.25	0.7586 ± 0.1033	48.8 ± 9.5
MLP	0.1009 ± 0.0040	0.0817 ± 0.0039	3.42 ± 0.18	0.8607 ± 0.0110	57.3 ± 6.1
Transformer	0.1018 ± 0.0099	0.0836 ± 0.0090	3.53 ± 0.39	0.8571 ± 0.0283	54.5 ± 13.0
Naive persistence	0.1024 ± 0.0000	0.0848 ± 0.0000	3.56 ± 0.00	0.8568 ± 0.0000	64.6 ± 0.0

At $h=1$, the average price error of MDHGNN approaches that of the parameter-free persistence baseline and outperforms the average RMSE values of GRU, MLP, and Transformer. At $h=2$, MLP and Naive persistence remain competitive; thus, the conclusion should emphasize the performance differences across various prediction steps.

VII. Ablation Experiment

The ablation experiment employed a masked ablation protocol: each random seed first trained the complete MDHGNN model, then during inference at the same weight settings, one hyper-edge channel or temporal path was masked.

variant	RMSE	R2	explain
Full MDHGNN	0.0685 ± 0.0018	0.9366 ± 0.0032	Complete the reasoning chain.
w/o Corr-HE (A1)	0.0687 ± 0.0035	0.9362 ± 0.0066	Shrink the correlation soft superedge.
w/o Domain-HE (A2)	0.0683 ± 0.0029	0.9368 ± 0.0054	Block prior hyperedges in the field.
w/o Adapt-HE (A3)	0.0683 ± 0.0029	0.9368 ± 0.0055	Hide adaptive hyperedges.
w/o GRU (A5)	0.1975 ± 0.0962	0.3484 ± 0.5106	Hide the GRU timing path.

Under the common true sample period, removing the GRU temporal path significantly degrades performance; the differences among the three types of super-edge channels are minimal, indicating a certain degree of redundancy in super-edge channels within shorter sample periods.

VIII. Operating Method

Installation dependencies:

```
pip install -r requirements.txt
```

Main Comparative Experiment:

```
python experiments/run_all.py --frequency weekly --horizons 1 2 --seeds 42 123 456
```

Ablation Experiment:

```
python experiments/run_masked_ablation.py --frequency weekly --horizon 1 --seeds 42 123 456
```

IX. Project Structure

File/Directory	explain
config.py	Centralize management of data, models, and training parameters.
main.py	Single-model training entry.
data/	Data reading, resampling, feature engineering, preprocessing, and datasets.
models/	Implementation of MDHGNN, supergraph convolution, prediction head, and baseline methods.
experiments/run_all.py	Conduct a main comparison experiment and generate result tables, DM tests, and charts.
experiments/run_masked_ablation.py	Screening ablation experiment.
experiments/redraw_publication_figures.py	Redraw the publication using the new checkpoint.
results/tables/	JSON metrics, DM validation, ablation summaries, and experiment manifest.
results/figures/	PNG, PDF, SVG charts.
tests/	unit test .